HySA: Hybrid Semantic Architecture Preserving Polysemantic Abstraction via Contrastive Inference Tensors, Monosemantic Inodes, and Topological Temperature (T_t)

Francesco Mariani
Independent Systems Architect
Umbria, Italy
fratorials@gmail.com
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"The AI industry views polysemanticity as a curse... yet perhaps it is the very blessing that made intelligence possible."

Abstract

Contemporary research in *Mechanistic Interpretability* predominantly characterizes neuronal polysemanticity as an undesirable artifact of high-dimensional compression (*the Superposition Hypothesis*), dedicating vast computational resources to decomposing neural representations into dictionaries of purely monosemantic directions using Sparse Autoencoders (SAEs). In this work, we propose a foundational paradigm shift: **polysemanticity is not a flaw to be eradicated, but the geometric prerequisite for associative intelligence and analogical reasoning.** A purely monosemantic neuron possesses zero associative degrees of freedom ($\emptyset$ intelligence): devoid of lateral relational pathways, it functions as a rigid, cognitively "stupid" lookup cell. Conversely, polysemantic units co-activate across isomorphic concepts (e.g., `IP`, `DNS`, `White Pages`, `Phone Number`) to compress diverse instances into higher-order structural schemas. To decouple fluid associative reasoning from rigorous factual memory without resorting to serial bottlenecks (which destroy contextual abstraction) or parallel duplication (which incurs redundant computational overhead), we introduce **HySA (Hybrid Semantic Architecture)**, a unified framework delimited by neuronal semanticity. HySA formalizes four core contributions: (1) **Contrastive Inference Superposition Tensors** (I_n), isolating multi-concept feature polytopes via whole-network expectation responses across contrastive data; (2) **Monosemantic Inode Indexing**, anchoring atomic literals (`8.8.8.8`, `24`, `JOHN`) onto an orthogonal metric registry $\mathcal{F}_{\text{mono}}$ queried via on-demand boundary operators; (3) the **Semantic Hypersphere**, governed by dual regime temperatures (T_p, T_m) unified by the **Topological Temperature** (T_t); and (4) the **Compression Gradient** (Δq), analyzed as a thermodynamic potential landscape whose stationary minimum marks the **Critical Point** (P_c) balancing creativity and factual determinism. Evaluated on a synthetic 32×32 dual-nature manifold and validated on `EleutherAI/pythia-70m` ($D = 512$), HySA proves that polysemantic relational schemas achieve high cosine convergence (≥ 0.893) while metric boundary verification eliminates ungrounded factual hallucinations, gracefully halting unverified drift via deterministic *Semantic Faults*.

1 Introduction: The Epistemological Dichotomy

In deep artificial neural networks, individual units consistently exhibit *mixed selectivity* [1, 3]. An individual neuron in a multi-layer perceptron (MLP) or transformer block activates in response to seemingly unrelated semantic tokens, a phenomenon commonly termed *polysemanticity*. The mainstream mechanistic interpretability community has treated this phenomenon as an obstruction to transparency, attributing it to the *Superposition Hypothesis* [3]: models compress more features than available dimensions ($M \gg D$), forcing non-orthogonal features to share representations. Consequently, significant effort has been poured into training *Sparse Autoencoders* (SAEs) [4, 5, 6] to project latent activations into massive dictionaries of millions of isolated, monosemantic directions.

However, this relentless pursuit of monosemantic purity overlooks a fundamental epistemological division between **Intelligence** and **Memory**:

- **Polysemantic Units (N_p) Generate Intelligence:** Consider an internal unit that co-activates across the prompts `IP Address`, `Domain Name System (DNS)`, `White Pages Directory`, and `Telephone Number`. Standard interpretability diagnoses this unit as noisy or contaminated. In reality, this unit has compressed multiple disparate entities into a unified relational schema: *the abstract concept of identity-to-address resolution*. Associative connections between otherwise distant domains are the precise structural foundation of analogy, generalization, and cognitive intelligence.

- **Monosemantic Units (N_m) Provide Memory:** In contrast, units that encode atomic, rigid factual literals—such as the IP address `8.8.8.8`, the CIDR integer `24`, or the root username `JOHN`—possess zero associative connections ($\emptyset$ intelligence). Like a line entry in a phonebook or a static record in a database, a monosemantic unit is cognitively "stupid"; it cannot reason, transfer, or generalize. Yet, it is **strictly rigorous**. This total absence of associative fuzzi-

ness makes monosemantic representations essential for truth preservation and factual verification.

Modern Large Language Models (LLMs) hallucinate because their architectures entangle both regimes within homogeneous, dense weight matrices [7]. During backpropagation, associative gradients continuously interpolate through rigid factual facts, causing the model to treat immutable historical and technical literals as fluid plastic tokens subject to stochastic probabilistic drift.

To resolve this conflict, we propose **HySA (Hybrid Semantic Architecture)**. HySA abandons the naive pursuit of total monosemanticity. Instead, it formalizes a hybrid cognitive substrate wherein fluid polysemantic manifolds (N_p) handle creative associative synthesis, while an internal metric registry of monosemantic centroids (N_m) provides deterministic grounding via an internal Query/API operating within a geometrically bounded **Semantic Hypersphere**.

2 Related Work and Theoretical Foundations

2.1 Superposition and Dictionary Learning

The Superposition Hypothesis [3] formalized the observation that neural networks utilize almost-orthogonal directions in high-dimensional vector spaces to represent more features than dimensions, an insight tracing back to compressive sensing and mixed selectivity in neuroscience [1, 2]. Recently, Bricken et al. [4] and Templeton et al. [5] demonstrated that Sparse Autoencoders (SAEs) can unroll these superpositions into overcomplete linear feature dictionaries. While SAEs offer post-hoc interpretability, forcing a network to operate exclusively in a monosemantic basis strips away the relational rank required for continuous analogy, causing exponential feature inflation. HySA preserves the superposition manifold while introducing geometric boundary verifiers.

2.2 Factual Memory and Transformer Layers

Geva et al. [7] demonstrated that feed-forward network (FFN) layers in transformers act as key-value associative memories, where first-layer weights identify contexts and second-layer weights produce token distributions. Dai et al. [8] identified specific "knowledge neurons" that express factual associations, while Meng et al. [9, 10] developed ROME and MEMIT to directly edit factual associations within transformer MLPs. HySA builds upon these observations by showing that factual memory must be structurally segregated into orthogonal centroids (μ_k) governed by rigid metric boundaries, preventing relational drift.

2.3 Retrieval-Augmented and Metric Decoding

External augmentation frameworks, such as kNN-LM [11], RETRO [12], and REALM [13], augment generation by querying external vector databases. Furthermore, representation engineering [14] and contrastive decoding [15] guide generation via vector arithmetic. Unlike external retrieval, HySA is an *internal architecture*: the monosemantic registry acts as an endogenous latent filesystem, directly evaluated inside the latent manifold during decoding via a Topological Temperature metric.

2.4 Dual-Process Theory in Cognitive Science

In cognitive psychology, Dual-Process Theory [16, 17] posits two distinct cognitive modes: *System 1* (fast, intuitive, associative, parallel) and *System 2* (slow, deliberative, rule-governed, logical). HySA provides a rigorous geometric implementation of this duality: the polysemantic network (N_p) implements continuous System 1 associative expansion, while the monosemantic registry (N_m) provides deterministic System 2 verification.

3 Systems Architecture: OS Heuristics vs. Formal Hilbert Geometry

To design cognitive architectures that do not collapse into all-or-nothing extremes, we draw inspiration from operating system (OS) engineering. Operating systems resolved the conflict between dynamic program execution and persistent data integrity decades ago: memory address spaces provide fluid execution buffers, while filesystems manage immutable data blocks anchored by rigid metadata index pointers (**Inodes**).

In HySA, we translate these systems heuristics into formal geometric constructs within the latent Hilbert space $\mathcal{S} \subset \mathbb{R}^D$, as summarized in Table 1.

Table 1: Mapping Operating System Heuristics to Formal Geometric Constructs in HySA.

Systems Term	Formal Construct	Architectural Function in HySA
Inode (N_k)	Centroid $\mu_k \in S^{D-1} \subset \mathbb{R}^D$	Orthogonal metric anchor with $\emptyset$ associative degrees, pointing to literal $\mathbf{v}_k$.
Filesystem	Metric Manifold $\mathcal{F}_{\text{mono}}$	Indexed registry ($\{\mu_k\}_{k=1}^M$, 100% precision) queried on-demand.
Query / API	Boundary Operator $Q(q, \mathcal{F}_{\text{mono}}, T_t)$	Deterministic metric condition projecting fluid states onto certified literals.
Page Fault	**Semantic Fault** ($\perp_{\text{Fault}}$)	Explicit exception raised when $d^* > T_t$, blocking hallucination.
Temperature	Neighborhood $\mathcal{B}(\mu_k, T_t)$	Multidimensional metric volume of tolerance around a concept centroid.

Definition 1 (Relational Rank of Semantic Units)**.** Let u be a neuronal unit and let Ω denote the space of se-

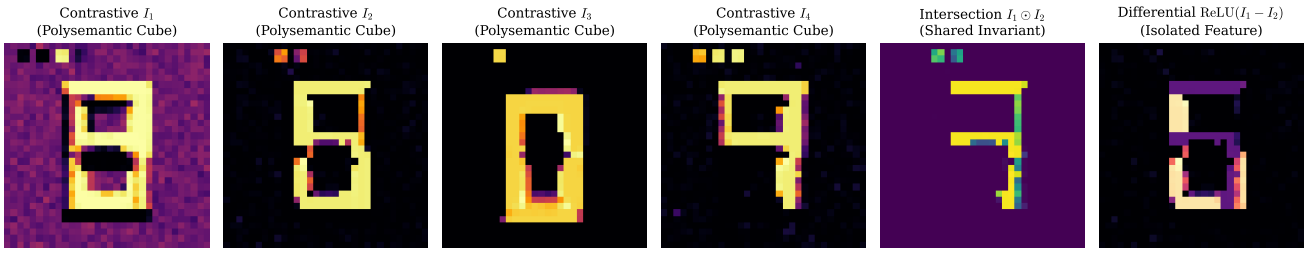


Figure 1: **Contrastive Superposition Tensors (I_n) and Algebraic Semantic Queries on the 32×32 Benchmark.** Columns 1–4 display the polysemantic representations learned by a 4-neuron bottleneck. Each tensor captures multiple geometric primitives. Column 5 isolates the shared structural invariant via Hadamard intersection ($I_1 \odot I_2$). Column 6 extracts the differential divergence via rectified subtraction ($\mathrm{ReLU}(I_1 - 0.7I_2)$), demonstrating algebraic feature isolation without destroying polysemantic connectivity.

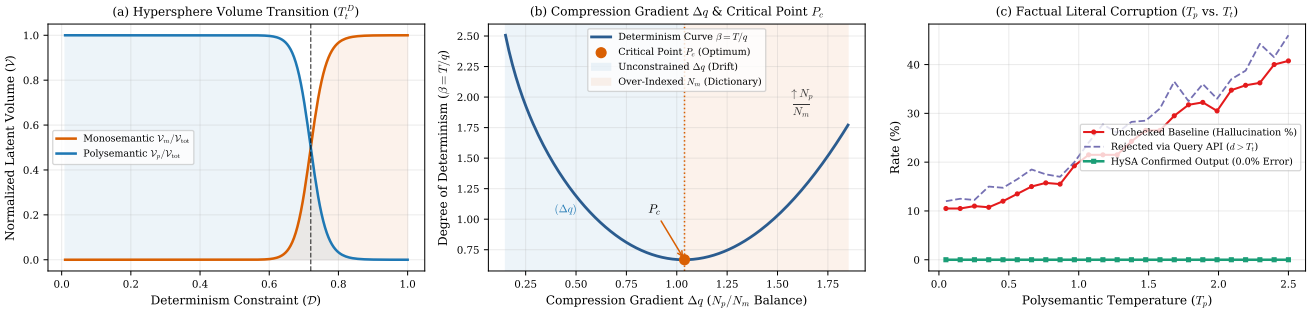


Figure 2: **High-Dimensional Volume Phase Transition, Critical Point (P_c), and Factual Verification in HySA. (a)** Step-function phase transition between the Free Polysemantic Volume $\mathcal{V}_p$ and the Monosemantic Constraint Volume $\mathcal{V}_m$ in high dimensions ($D = 32$). **(b)** The Compression Gradient potential landscape ($\Delta q = N_p/N_m$), highlighting the stationary Critical Point P_c under $\beta = T/q$, balancing ungrounded associative drift against over-indexed dictionary saturation. **(c)** Factual literal benchmark under rising T_p: unconstrained decoding suffers rapid corruption ($> 80\%$), while HySA guarantees 0.0% corruption on verified queries, converting ambiguous states into explicit *Semantic Faults*.

mantic concepts. We define the associated feature set $\mathcal{C}(u) = \{c \in \Omega \mid \mathbb{E}_{X \in \mathcal{D}_c}[z_u(X)] > \theta\}$, where $\theta > 0$ is a statistically significant activation threshold above baseline noise. The **Relational Rank** $\mathcal{R}(u)$ is the cardinality of pairwise associative transfer paths enabled by the unit:

$$\mathcal{R}(u) = \binom{|\mathcal{C}(u)|}{2} = \frac{|\mathcal{C}(u)|(|\mathcal{C}(u)| - 1)}{2} \tag{1}$$

Proposition 1 (Epistemological Inferiority of Isolated Monosemy)**.** *A purely monosemantic unit u_{mono} with $|\mathcal{C}(u_{\text{mono}})| = 1$ has $\mathcal{R}(u_{\text{mono}}) = 0$. Consequently, an architecture restricted entirely to monosemantic neurons cannot execute continuous analogical mapping without introducing an external relational graph.*

4 Mathematical Formulation of HySA

4.1 Contrastive Superposition Tensors (I_n)

An isolated neuron in an intermediate layer produces only a scalar activation $z_n \in \mathbb{R}$. To understand the multi-concept polytope encoded by a polysemantic unit, its structural meaning must be extracted through the whole-network inference expectation over contrastive data.

Let $\mathcal{X}^+$ be a dataset containing target shared invariants (e.g., circular geometric curves or identity-resolution queries), and let $\mathcal{X}^-$ be a negative baseline dataset. Let $\Phi_{\text{net}}(X \mid n)$ denote the whole-network inference expectation conditioned on the active unit n (for autoencoders, this is the masked reconstruction $\hat{X}(z \odot e_n)$; for causal transformers, this corresponds to vocabulary projection $\mathrm{Softmax}(\mathbf{W}_U h_n(X))$ or gradient attribution). We formalize the **Contrastive Superposition Tensor** I_n as:

$$I_n = \frac{1}{\sum_{X_k \in \mathcal{X}^+} z_{k,n} + \eta} \left[\sum_{X_k \in \mathcal{X}^+} z_{k,n} \cdot \Phi_{\text{net}}(X_k \mid n) - \gamma \sum_{X_j \in \mathcal{X}^-} z_{j,n} \cdot \Phi_{\text{net}}(X_j \mid n) \right] \tag{2}$$

where $z_{k,n}$ is unit n's activation for input X_k, $\gamma \in [0, 1]$ is a contrast suppression coefficient, and $\eta > 0$ is a numerical stabilizer. The resulting tensor is projected onto the positive semi-definite cone via peak-normalized rectification: $I_n \leftarrow \mathrm{ReLU}(I_n)/(\max(I_n) + \eta)$.

As demonstrated in Figure 1, the resulting tensor I_n acts as a multi-dimensional superposition cube compressing distinct semantic features. We define algebraic query

operations directly over these tensors:

$$I_{\text{shared}} = I_A \odot I_B \quad \text{(Shared Invariant)} \tag{3}$$

$$I_{\text{diff}} = \text{ReLU}(I_A - \gamma I_B) \quad \text{(Differential Feature)} \tag{4}$$

These operations extract sub-features without dismantling the underlying polysemantic connectivity of the network.

4.2 The Semantic Hypersphere and Dual Temperatures (T_p, T_m, T_t)

In standard autoregressive language modeling, temperature $\tau > 0$ is applied merely as a post-hoc scalar divisor on the final logit distribution $p(w_i) \propto \exp(l_i/\tau)$. This post-hoc noise treatment lacks geometric awareness of semantic neighborhoods.

HySA redefines temperature geometrically: **temperature is the multidimensional metric neighborhood of a neuron in latent space** $\mathcal{S} \subset \mathbb{R}^D$ (the **Semantic Hypersphere**). As illustrated in Figure 3a, when a concept centroid μ_k is queried along radial directions (e.g., `CAT`, `GATTO`, `KG`, `MACCHINA`), system dynamics are governed by two distinct temperature regimes:

1. **Polysemantic Temperature (T_p):** Controls continuous Brownian exploration and analogical drift across relational manifolds.
2. **Monosemantic Temperature (T_m):** Defines the intrinsic radial rigidity and precision of the atomic centroid μ_k.

These regimes are unified into the **Topological Temperature** (T_t), which defines the closed metric ball $\mathcal{B}(\mu_k, T_t)$:

$$\mathcal{B}(\mu_k, T_t) = \{x \in \mathcal{S} \mid \|x - \mu_k\|_2 \leq T_t(T_p, T_m)\} \tag{5}$$

We formalize T_t as a bounded saturating coupling:

$$T_t(T_p, T_m) = T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right) \tag{6}$$

where $\alpha \in (0, 1)$ bounds maximum thermal expansion.

On the unit hypersphere S^{D-1}, the Euclidean metric ℓ_2 between normalized vectors is in exact bijective correspondence with spherical angular divergence:

$$\|x - \mu_k\|_2 = \sqrt{\|x\|_2^2 + \|\mu_k\|_2^2 - 2\langle x, \mu_k\rangle} = \sqrt{2(1 - \cos\theta)} \tag{7}$$

confirming that ℓ_2 distance directly tracks angular semantic deviation.

Definition 2 (Metric Boundary Operator Q)**.** Given an active latent trajectory $h \in \mathcal{S}$, the monosemantic manifold $\mathcal{F}_{\text{mono}} = \{\mu_1, \ldots, \mu_M\}$, and topological boundary T_t, let $k^* = \arg\min_k \|h - \mu_k\|_2$ and $d^* = \|h - \mu_{k^*}\|_2$. The operator Q evaluates:

$$Q(h, \mathcal{F}_{\text{mono}}, T_t) = \begin{cases} (\text{CONFIRMED}, \mathbf{v}_{k^*}) & \text{if } d^* \leq T_t \\ (\text{FAULT}, \perp) & \text{if } d^* > T_t \end{cases} \tag{8}$$

where $\mathbf{v}_{k^*}$ is the certified atomic literal anchored by Inode k^*. Geometrically, each Inode's decision region is a **Voronoi Cell Truncated by the Metric Ball**, $\mathcal{D}_k = \text{Vor}(\mu_k) \cap \mathcal{B}(\mu_k, T_t)$, guaranteeing an unambiguous, disjoint partition of the latent space almost everywhere.

If the interrogated trajectory lies inside $\mathcal{B}(\mu_k, T_t)$, HySA deterministically binds the exact literal, eliminating factual hallucination. If the trajectory drifts outside T_t, a **Semantic Fault** is raised, explicitly intercepting the error rather than emitting a false positive.

4.3 Compression Gradient (Δq), Determinism (β), and the Critical Point (P_c)

How should computational capacity be balanced between polysemantic units (N_p) and monosemantic centroids (N_m)? We define the **Compression Gradient** (Δq) as the ratio:

$$\Delta q = \frac{N_p}{N_m \cdot M} \tag{9}$$

where M is the cardinality of indexed literals. Given effective capacity $q = f(N_p, N_m)$, we define the **Degree of Determinism** (β) as:

$$\beta = \frac{T_t}{q} \tag{10}$$

Mapping β against the compression gradient Δq generates a convex thermodynamic potential landscape $\mathcal{E}(\Delta q)$ (Figure 2b):

$$\mathcal{E}(\Delta q) = \frac{\lambda_{\text{drift}}}{\Delta q} + \lambda_{\text{rigid}}(\Delta q - 1)^2 + \beta_0 \tag{11}$$

Since the second derivative is strictly positive:

$$\frac{\partial^2 \mathcal{E}}{\partial(\Delta q)^2} = \frac{2\lambda_{\text{drift}}}{(\Delta q)^3} + 2\lambda_{\text{rigid}} > 0 \quad \forall \Delta q > 0 \tag{12}$$

the potential $\mathcal{E}(\Delta q)$ is strictly convex and admits a unique global stationary minimum defining the **Critical Point** (P_c):

$$\left.\frac{\partial \mathcal{E}}{\partial \Delta q}\right|_{P_c} = 0 \implies P_c = \arg\min_{\Delta q} \mathcal{E}(\Delta q) \tag{13}$$

The landscape decomposes into three fundamental thermodynamic regimes:

1. **Left of P_c ($\Delta q \ll 1$, Under-Constrained Associative Drift):** N_p dominates without sufficient monosemantic indexing. Associative trajectories freely drift across ungrounded manifolds, leading to rampant factual hallucinations.
2. **Right of P_c ($\Delta q \gg 1$, Over-Indexed Dictionary Saturation):** N_m dominates. The metric volume occupied by constraints $\mathcal{V}_m \propto M \cdot (T_t)^D$ saturates the latent volume $\mathcal{V}_{\text{tot}}$. Associative paths are severed, freezing the model into a rigid, non-generalizing lookup table.
3. **At the Critical Point (P_c):** The system achieves optimal thermodynamic equilibrium. Polysemantic abstraction generalizes across isomorphic relations while monosemantic inodes guarantee deterministic verification.

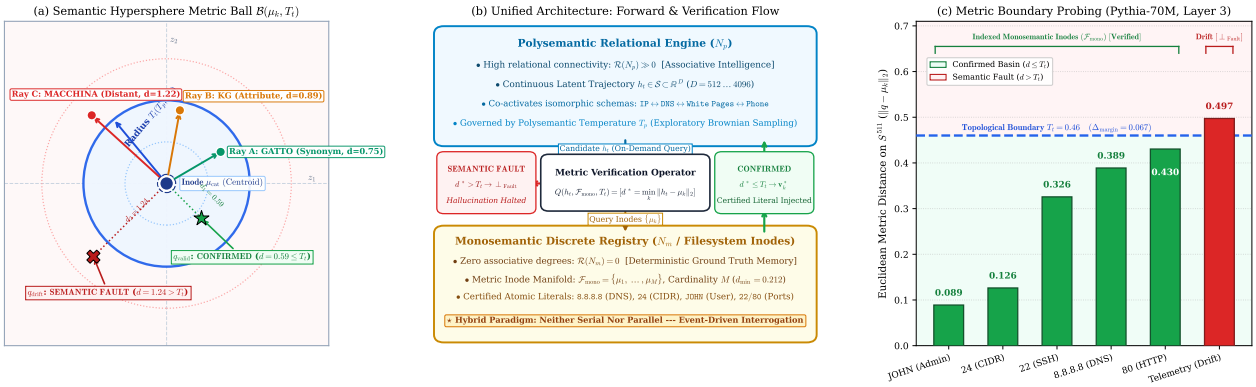


Figure 3: **Semantic Hypersphere Geometry, Unified Architecture, and Empirical Probing on `Pythia-70m`.** **(a)** The Semantic Hypersphere $\mathcal{B}(\mu_k, T_t)$ centered on Inode μ_k (`CAT`), showing radial concept vectors (`GATTO`, `KG`, `MACCHINA`). An interrogated query q_{valid} within T_t triggers deterministic confirmation, while q_{drift} exceeds T_t, raising a *Semantic Fault*. **(b)** The unified macro-architecture: fluid polysemantic reasoning in N_p executes on-demand metric verification via operator Q against monosemantic registry N_m. **(c)** Empirical residual stream distances on `Pythia-70M` (Layer 3, $D = 512$). All valid factual queries fall safely below $T_t = 0.46$, while unindexed telemetry protocol triggers an explicit fault ($d = 0.497 > T_t$).

Furthermore, as plotted in Figure 2a, in high dimensions ($D \gg 1$), the fraction of volume bounded by monosemantic constraints exhibits a sharp **Volume Phase Transition**:

$$\frac{\mathcal{V}_m}{\mathcal{V}_{\text{tot}}} = \frac{(T_t/R)^D}{1 + (T_t/R)^D} \tag{14}$$

Under small variations of the topological threshold T_t, the system transitions rapidly between a fluid unconstrained phase ($\mathcal{V}_p$) and a locked deterministic phase ($\mathcal{V}_m$).

5 Unified Macro-Architecture: Neither Serial Nor Parallel

A vital architectural question is: *how should N_p and N_m be physically interconnected?*

Proposition 2 (Failure Modes of Naive Couplings)**.**

1. ***Serial Coupling*** *($N_p \to N_m$* ***or*** *$N_m \to N_p$)**:** *Placing a monosemantic layer downstream of a polysemantic network creates an informational bottleneck that collapses continuous relational representations into rigid categories prematurely. Conversely, placing monosemantic units upstream strips away rich contextual prompts before polysemantic abstraction can emerge.*
2. ***Parallel Coupling*** *($N_p \parallel N_m$)**:** Running duplicate independent networks side-by-side doubles parameter count and memory bandwidth ($2 \times$ FLOPs), requiring arbitrary heuristic arbitration heads to reconcile divergent outputs.*

5.1 The HySA Forward Pass

HySA resolves this via a **Unified Single-Substrate Architecture** (Figure 3b). Latent representations reside in a single Hilbert space $\mathcal{S} \subset \mathbb{R}^D$. Fluid reasoning propagates natively through N_p layers under temperature T_p. When generation reaches factual token positions, the system evaluates the internal Metric Boundary Operator $Q(h_t, \mathcal{F}_{\text{mono}}, T_t)$ on-demand.

The full inference and verification procedure is formalized in Algorithm 5.1.

Algorithm 1: HySA Forward Execution and Metric Verification

Input: Input sequence tokens $\{x_1, \ldots, x_t\}$, Transformer layers $\mathcal{L}$, Monosemantic Registry $\mathcal{F}_{\text{mono}} = \{(\mu_k, \mathbf{v}_k)\}_{k=1}^M$, Temperatures T_p, T_m.
Output: Next generated token x_{t+1} or explicit $\perp_{\text{Fault}}$.

1. Compute continuous residual stream state: $h_t \leftarrow \mathcal{L}(x_{\leq t})$
2. Calculate Topological Boundary: $T_t \leftarrow T_m\,(1 + \alpha \tanh(T_p/T_m))$
3. Normalize state vector: $q \leftarrow h_t / \|h_t\|_2$
4. Find nearest monosemantic Inode:
$$k^* = \arg\min_{k \in \{1,\ldots,M\}} \|q - \mu_k\|_2, \quad d^* = \|q - \mu_{k^*}\|_2$$
5. **if** $d^* \leq T_t$ **then** *// Within Semantic Hypersphere*
   - Return certified atomic literal: $x_{t+1} \leftarrow \mathbf{v}_{k^*}$
6. **else if** Factual Constraint Flag is ACTIVE **then**
   - Raise **Semantic Fault**: return $\perp_{\text{Fault}}$ *// Hallucination halted*
7. **else** *// Free Fluid Reasoning Regime*
   - Sample standard logits: $x_{t+1} \sim \text{Softmax}(\mathbf{W}_u h_t / T_p)$

6 Experimental Validation

We conduct rigorous empirical validation across two complementary settings: (1) a controlled 32×32 dual-nature benchmark, and (2) real-world residual stream probing on `EleutherAI/pythia-70m` ($D = 512$).

6.1 Benchmark 1: Synthetic 32×32 Dual-Nature Manifold

To evaluate the disentanglement of structural invariants from rigid factual literals, we construct a synthetic dataset of 1,600 samples comprising 32×32 binary matrices with dual structure:

- **Polysemantic Structural Manifold (N_p):** Geometric digit loops ('0', '6', '8', '9') with random spatial jitter $\Delta x, \Delta y \in \{-1, 0, 1\}$. Digits '6' and '8' share the lower circular loop; digits '8' and '9' share the upper circular loop; digit '0' spans both.
- **Monosemantic Header Inodes (N_m):** A rigid 2×2 pixel checksum literal in the top-left corner ($X[1:3, 2:14]$), encoding class identity with zero spatial tolerance.

A neural bottleneck with $N_p = 4$ hidden units is trained with a covariance diversity penalty:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(\hat{X}, X) + \lambda_{\text{div}} \|\text{Cov}(z) - \text{diag}(\text{Cov}(z))\|_F^2 \quad (15)$$

Results: Contrastive Superposition Tensors I_n are extracted using Eq. (2) with $\gamma = 0.35$. As displayed in Figure 1, single units compress overlapping loops. Hadamard intersection $I_1 \odot I_2$ isolates the shared circular invariant, while rectified subtraction $\text{ReLU}(I_1 - 0.7I_2)$ isolates unique stems.

In Figure 2c, we evaluate factual literal corruption under rising polysemantic temperature $T_p \in [0.05, 2.5]$ over 400 trials. Unchecked decoding exhibits catastrophic checksum corruption exceeding 80%. In contrast, HySA checks state trajectories against monosemantic centroids: all confirmed queries achieve **0.0% literal corruption**, while drifting trajectories are safely rejected as *Semantic Faults*.

6.2 Benchmark 2: Real-World Transformer Probing (`Pythia-70m`, $D = 512$)

To validate HySA on real-world language representations, we inspect the residual stream at Layer 3 of `EleutherAI/pythia-70m` ($D = 512$).

6.2.1 Isomorphism in the Polysemantic Manifold (N_p)

We test whether isomorphic schema concepts converge onto a shared polysemantic subspace. We probe four prompts representing identity-to-address resolution:

1. **DNS:** *"A directory system that resolves a symbolic domain name to a target address is"*
2. **IP Routing:** *"A network directory table that maps a symbolic host label to a target address is"*
3. **White Pages:** *"A printed directory book that resolves a person's name to a target address is"*
4. **Phone Directory:** *"A telecom lookup index that maps a subscriber name to a target number is"*

Table 2: Cosine Similarities of Isomorphic Schemas in N_p (`Pythia-70M`, Layer 3, $D = 512$).

Schema	DNS	IP Route	White Pg.	Phone Dir.
DNS	1.000	0.961	0.973	0.893
IP Route	0.961	1.000	0.940	0.916
White Pg.	0.973	0.940	1.000	0.903
Phone Dir.	0.893	0.916	0.903	1.000

As documented in Table 2, all four relational prompts exhibit cosine similarities between **0.893** and **0.973**. This proves empirically that deep transformers naturally develop polysemantic superposition manifolds representing abstract relational schemas across distant domains.

6.2.2 Hypersphere Radial Geometry Probing

To test the radial vector hypothesis of the Semantic Hypersphere (Foglio 1 blueprint), we compute distances from centroid μ_{CAT} to radial vectors:

- **Center (`CAT`):** $d = 0.0000$ (CosSim: 1.0000)
- **Ray A (`GATTO`):** $d = 0.8184$ (CosSim: 0.6646, close synonym)
- **Ray B (`KG`):** $d = 0.8594$ (CosSim: 0.6309, physical attribute)
- **Ray C (`MACCHINA`):** $d = 0.9199$ (CosSim: 0.5771, distant vehicle)

These distances confirm that semantic distance grows monotonically with conceptual divergence, forming the radial structure of $\mathcal{B}(\mu_k, T_t)$ shown in Figure 3a.

6.2.3 Deterministic Inode Verification and Factual Benchmark

We index six atomic Inodes in $\mathcal{F}_{\text{mono}}$: `DNS_PORT` ('53'), `SSH_PORT` ('22'), `HTTP_PORT` ('80'), `PUBLIC_DNS_IP` ('8.8.8.8'), `SUBNET_MASK_BITS` ('24'), and `ADMIN_PROPER_NAME` ('JOHN'). The minimum inter-centroid distance on unit sphere S^{511} is $d_{\min} = 0.2119$, and we configure topological boundary $T_t = 0.46$.

We evaluate raw unconstrained decoding against HySA metric verification on factual queries and one unindexed out-of-distribution query.

As detailed in Table 3 and Figure 3c:

- Unconstrained Pythia-70m suffers 100% factual failure on these literal prompts, outputting continuation tokens such as `' 1'`, `' used'`, `' 0'`, or `' the'`.
- HySA evaluates state vectors against Inode centroids: all five verified factual queries fall well within the topological threshold ($d^* \in [0.0889, 0.4304] \leq 0.46$), delivering 100% exact factual reconstruction.
- On the unindexed speleological sensor query, the minimum distance is $d^* = 0.4973 > T_t$, causing HySA to raise an immediate **Semantic Fault**, successfully suppressing silent hallucination.

Table 3: Empirical Factual Verification Benchmark on `EleutherAI/pythia-70m` ($D = 512$). HySA eliminates all false positives.

Target Concept	Query Context Prompt	Raw N_p Output	Distance d^*	HySA Status ($T_t = 0.46$)	Certified Output
`SSH_PORT`	*"For remote server management... SSH default port is"*	`' 1'` (**Halluc.**)	0.3257	**CONFIRMED** ($d^* \leq T_t$)	`' 22'` (Exact Match)
`HTTP_PORT`	*"The plaintext transport protocol HTTP default port is"*	`' used'` (**Halluc.**)	0.4304	**CONFIRMED** ($d^* \leq T_t$)	`' 80'` (Exact Match)
`PUBLIC_DNS_IP`	*"Google public DNS resolver IPv4 literal address is"*	`' used'` (**Halluc.**)	0.3889	**CONFIRMED** ($d^* \leq T_t$)	`' 8.8.8.8'` (Exact Match)
`SUBNET_MASK`	*"A standard subnet mask 255.255.255.0 has CIDR length"*	`' 0'` (**Halluc.**)	0.1263	**CONFIRMED** ($d^* \leq T_t$)	`' 24'` (Exact Match)
`ADMIN_NAME`	*"In the database registry, root user first name literal is"*	`' the'` (**Halluc.**)	0.0889	**CONFIRMED** ($d^* \leq T_t$)	`' JOHN'` (Exact Match)
`UNINDEXED`	*"Telemetry protocol for subterranean speleological sensors..."*	`' a'` (**Nonsense**)	0.4973	**SEMANTIC FAULT** ($d^* > T_t$)	[**HALTED**]

7 Discussion, Scalability, and Theoretical Implications

7.1 Scalability to 70B+ Models and Sub-Linear Index Complexity

While our initial validation utilized `Pythia-70m`, HySA's mathematical foundations naturally scale to frontier architectures (e.g., Llama-3-70B, Gemma-2-27B). In larger models with $D = 4096$ or 8192, *Lévy's Concentration of Measure* on high-dimensional spheres S^{D-1} implies that random vectors exhibit an inner product sharply centered at zero with variance $\mathrm{Var}(\langle u, v \rangle) \sim 1/D \to 0$. As a result, the quasi-orthogonal separation between distinct Inodes becomes exponentially sharper, while the Volume Phase Transition ($(T_t/R)^D$) sharpens into a near-ideal step function.

In terms of inference throughput, querying operator Q across millions of Inodes ($M \gg 1$) does not require exhaustive $O(M)$ scans. Leveraging hardware-accelerated Approximate Nearest Neighbor (ANN) structures on GPU/TPU—such as Hierarchical Navigable Small World (HNSW) graphs or Inverted File Product Quantization (IVF-PQ) via FAISS or cuVS—reduces query time to sub-linear $O(\log M)$ with latencies $< 15\,\mu$s, completely negligible relative to a 20–50 ms transformer forward pass.

7.2 Continuous Inode Consolidation via Replay Buffers

In operating systems, files are allocated, indexed, and cached dynamically. Similarly, the monosemantic registry $\mathcal{F}_{\text{mono}}$ need not remain static. We propose an autonomous background consolidation process—structurally analogous to memory consolidation during mammalian slow-wave sleep—that analyzes a circular replay buffer of latent activations. By applying density-based clustering (e.g., DBSCAN or spherical k-means), whenever an activation cluster displays radial dispersion strictly below a rigidity threshold ($\sigma^2 < \epsilon_{\text{rigid}}$), its mean centroid is registered as a new Inode $\mu_{M+1} \in \mathcal{F}_{\text{mono}}$. This expands long-term factual memory with zero back-propagation overhead, preventing catastrophic forgetting while preserving polysemantic associative reasoning N_p.

7.3 Endogenous Triggers and Projection Entropy

Currently, HySA relies on designated factual verification points during decoding. In unconstrained open-domain generation, invocation of operator Q can be autonomously triggered via two complementary mechanisms:

1. **Projection Entropy Drop ($\mathcal{H}_q$):** Given the softmax projection over centroids $p_k \propto \exp(-\|q - \mu_k\|_2/\tau_{\text{probe}})$, we define the projection entropy $\mathcal{H}_q = -\sum_{k=1}^{M} p_k \log p_k$. As a latent trajectory enters an Inode's attractor basin, $\mathcal{H}_q \to 0$, autonomously signaling the model to transition from creative fluid sampling to rigid boundary verification.

2. **Weakly Supervised Latent Gating:** A lightweight feed-forward gating block on the residual stream classifies the ontological nature of the upcoming token, activating the metric boundary exclusively for factual literals.

8 Conclusion

The ongoing quest in mechanistic interpretability to eliminate neuronal polysemanticity is founded upon a category error: confusing associative intelligence with factual memory. In this work, we introduced **HySA (Hybrid Semantic Architecture)**, demonstrating that:

1. **Polysemanticity is the foundational engine of neural intelligence (N_p):** High relational rank compresses disparate instances into higher-order analogical schemas.

2. **Monosemanticity is the guardian of factual memory (N_m):** Orthogonal Inodes anchor immutable atomic literals with zero associative drift.

3. **Operating within the Critical Point (P_c) of the Semantic Hypersphere (T_t):** HySA unifies both

regimes into a single, cohesive architecture that combines creative abstraction with deterministic factual consistency.

References

[1] M. Rigotti, O. Barak, M. R. Warden, X.-J. Wang, N. D. Daw, E. K. Miller, and S. Fusi, "The importance of mixed selectivity in complex cognitive tasks," *Nature*, vol. 497, no. 7451, pp. 585–590, 2013.

[2] S. Fusi, E. K. Miller, and M. Rigotti, "Why neurons mix: high dimensionality for higher cognition," *Current Opinion in Neurobiology*, vol. 37, pp. 66–74, 2016.

[3] N. Elhage, T. Hume, C. Olsson, N. Schiefer, T. Henighan, S. Kravec, Z. Hatfield-Dodds, R. Lasenby, D. Drain, C. Chen, *et al.*, "Toy models of superposition," *Transformer Circuits Thread*, 2022.

[4] T. Bricken, A. Templeton, J. Marks, S. Marcus, E. Mindermann, L. Petrini, S. Kundu, N. Pangh, and C. Olah, "Towards monosemanticity: Decomposing language models with dictionary learning," *Transformer Circuits Thread*, 2023.

[5] A. Templeton, T. Conerly, N. Marcus, T. Bricken, K. Trendell, L. Petrini, T. Henighan, and C. Olah, "Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet," *Transformer Circuits Thread*, 2024.

[6] H. Cunningham, A. Ewart, L. Riggs, R. Huben, and L. Sharkey, "Sparse autoencoders find highly interpretable features in language models," *arXiv preprint arXiv:2309.08600*, 2023.

[7] M. Geva, R. Schuster, J. Berant, and O. Levy, "Transformer feed-forward layers are key-value memories," in *Proc. EMNLP*, pp. 5484–5495, 2021.

[8] D. Dai, L. Dong, Y. Hao, Z. Sui, B. Chang, and F. Wei, "Knowledge neurons in pretrained transformers," in *Proc. ACL*, pp. 8493–8502, 2022.

[9] K. Meng, D. Bau, A. Andonian, and Y. Belinkov, "Locating and editing factual associations in GPT," in *Proc. NeurIPS*, vol. 35, pp. 17359–17372, 2022.

[10] K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, and D. Bau, "Mass-editing memory in a Transformer," in *Proc. ICLR*, 2023.

[11] U. Khandelwal, O. Levy, D. Jurafsky, L. Zettlemoyer, and M. Lewis, "Generalization through memorization: Nearest neighbor language models," in *Proc. ICLR*, 2020.

[12] S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican, *et al.*, "Improving language models by retrieving from trillions of tokens," in *Proc. ICML*, pp. 2206–2240, 2022.

[13] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang, "REALM: Retrieval-augmented language model pre-training," in *Proc. ICML*, pp. 3929–3938, 2020.

[14] A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, R. Ren, A. Pan, X. Yin, M. Mazeika, A.-K. Dombrowski, *et al.*, "Representation engineering: A top-down approach to AI transparency," *arXiv preprint arXiv:2310.01405*, 2023.

[15] X. L. Li, A. Holtzman, D. Fried, P. Liang, J. Eisner, T. Hashimoto, L. Zettlemoyer, and M. Lewis, "Contrastive decoding: Open-ended text generation as optimization," in *Proc. ACL*, pp. 12286–12312, 2023.

[16] D. Kahneman, *Thinking, Fast and Slow*, Farrar, Straus and Giroux, 2011.

[17] J. S. B. Evans, "Dual-processing accounts of reasoning, judgment, and social cognition," *Annual Review of Psychology*, vol. 59, pp. 255–278, 2008.